Enumerating Solid Sudoku Subcubes

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Abstract

In this paper, we enumerate and bound subcubes of the Solid Sudoku Cube. We establish that there is 1 regular $2 \times 2 \times 2$ subcube and 448 regular $3 \times 3 \times 3$ subcubes, unique up to relabeling. Furthermore, using matrix permanents, we derive generalized combinatorial formulas that bound the total number of valid configurations for any arbitrary $x \times y \times z$ subcube.

1 Introduction

A Sudoku grid is a 9×9 Latin square divided into nine 3×3 regions with the condition that every column, row and region is filled with the digits $1-9$, each of the digits appearing only once.

Three dimensionsional Sudoku has been constructed in a number of different ways. Church (2005) constructed a Sudoku Cube by layering nine Sudoku grids on top of each other with the added structure that three adjacent faces also formed Sudoku grids. Lambert & Whitlock (2010) generalized the Dion Cube so that all six faces could be used as Sudoku grids.

Solid Sudoku Cubes (SSCs) were introduced in Najafian *et al*. (2018). An order n SSC is a three-dimensional $n^2 \times n^2 \times n^2$ array constrained such that every integer from 1 to n^2 appears once along every row, column, and depth axis, with each orthogonal slice structurally functioning as a valid two-dimensional Sudoku grid. We discuss two classes of subcubes of the SSC;

- The Regular Subcube: An $n \times n \times n$ volume that perfectly aligns with the orthogonal slice constraints of the parent SSC. Structurally, a regular subcube is

An interactive, playable model of the order 3 regular subcube puzzle can be accessed `https://xfpqc.github.io/Cubiku`.

equivalent to a continuous sequence of n perfectly aligned planar slices (which we denote as blocks $B_1, \ldots, B_n$).

- The Arbitrary Subcube: Any $x \times y \times z$ volumetric block. This includes irregular geometries, as well as full $n \times n \times n$ subcubes that are placed such that their boundaries are offset from the natural block slices of the parent grid.

The problem of enumerating all valid Sudoku grids was resolved by Felgenhauer & Jarvis (2005). Russell & Jarvis (2006) found the number of essentially different, nonequivalent Sudoku grids.

1.1 Definitions and Theorems

Definition 1.1. Let $A = (a_{ij})$ be an $m \times n$ matrix over any commutative ring with $m \leq n$. Let $P(m,n)$ denote the set of all one-to-one functions from $\{1,2,\ldots,m\}$ to $\{1,2,\ldots,n\}$. The *permanent* of A, $\mathrm{per}(A)$, is defined by

$$\mathrm{per}(A) = \sum_{\sigma \in P(m,n)} a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{m,\sigma(m)}$$

Definition 1.2. Let X_1,X_2,...,X_m be subsets, not necessarily distinct, of an n-set $S = \{x_1, x_2, ..., x_n\}$. Let $A = (a_{ij})$ be the $(0,1)$-matrix whose entries are defined as follows:

$$a_{ij} = \begin{cases} 1 & \text{if} x_j \in X_i \\ 0 & \text{if} x_j \notin X_i \end{cases}$$

The matrix A is called the *incidence matrix* for the subsets X_1,X_2,...,X_m of the n-set S.

Definition 1.3. Let X_1,X_2,...,X_m be subsets of an n-set S. A sequence $(s_1, s_2, ..., s_m)$ of m distinct elements of S is said to form a *system of distinct representatives* (SDR) if $s_i \in X_i$, $i = 1, ..., m$.

(Minc (1984)) The number of SDRs for a set of sets with incidence matrix M is $\mathrm{per}(M)$.

Definition 1.4. Λ_n^k is the set of $n \times n$ $(0,1)$-matrices in which each row and column sum to k.

Theorem 1.1 (Minimum values of per(A), $A \in \Lambda_n^k$)**.** *(Wanless (2006)) For any integers $n \geq k \geq 1$ and any $A \in \Lambda_n^k$,*

$$\text{per}(A) > (\frac{(k-1)^{k-1}}{k^{k-2}})^n$$

Theorem 1.2 (Maximum values of per(A), $A \in \Lambda_n^k$)**.** *(Brègman (1973), Wanless (2003)) For $n = tk + r$ with $0 \leq r < k$*

$$(k!)^t r! \leq \max_{A \in \Lambda_n^k} \text{per}(A) \leq (k!)^{\frac{n}{k}}$$

Theorem 1.3 (Brègman-Minc Theorem)**.** *(Schrijver (1978)) For an $n \times n$ $(0,1)$-matrix A, let r_i denote the row sum in the i-th row. The permanent of A, denoted* per$A$*, satisfies:*

$$\text{per}\, A \leq \prod_{i=1}^{n} (r_i!)^{1/r_i}$$

Lemma 1.1. (Borovskikh & Korolyuk (2020)) Let A be an $m \times n$ matrix where $m \leq n$. Let E be an $(n-m) \times n$ matrix in which all entries are 1. If $\bar{A}$ is the square $n \times n$ augmented matrix defined by,

$$\bar{A} = \begin{bmatrix} A \\ E \end{bmatrix}$$

then the permanent of the augmented matrix $\bar{A}$ is given by:

$$per(\bar{A}) = (n-m)!per(A)$$

A regular $n \times n \times n$ subcube is equivalent to the top n blocks (denoted $B_1, \ldots, B_n$) of an n-Sudoku grid, with the additional constraint that similar columns across these blocks must collectively contain the digits $\{1, \ldots, n^2\}$ once. Because of this structure, a solid subcube of depth z naturally spans z consecutive blocks. When a digit is assigned to a cell in B_m, the column constraint immediately removes that digit from the available pool for the corresponding cell in B_{m+1}. Consequently, the three-dimensional depth rules are preserved throughout the two-dimensional presentation.

2 Bounds for Regular $n \times n \times n$ Subcubes

2.1 Counting Regular Subcubes of order 2

Exchanging any two digits in a valid grid results in another valid grid. Any grid can be relabeled so that the leftmost block (B_1) is given by

1	2		
3	4		

In this case we say the grid is in standard form, and our count reduces by $n!$. Our problem reduces to counting the number of cubes with B_1 in standard form.

Lemma 2.1. Up to standard-form relabeling, there exists a unique valid regular subcube configuration of order 2.

Proof. Assume B_1 is fixed in standard form. For the subsequent block B_2, the available candidates for any cell (r,c) are bounded by the elements already placed in the corresponding row and column of B_1. Since every digit must appear once in each slice, these cross-slice constraints leaves one valid digit placement per cell. Therefore, B_2 is uniquely determined. □

There is only one way to complete the grid that satisfies the constraints and so, with our ordered entries, we have 1 regular subcube of order 2 up to relabeling, and $4! \times 1 = 24$ valid regular subcubes.

2.2 Counting Regular Subcubes of order 3

As before, we do relabeling on B_1, reducing our counting by $9! = 362,880$. We now have B_1 in standard form.

Determining possible entry combinations is essentially an SDR enumeration problem. Each of the nine cells in B_2 corresponds to a specific subset of the set $\{1,2,\ldots,9\}$, and we must pick distinct representatives from each of these subsets to form a valid block.

We therefore need to count the number of ways we can fill B_2. We have the following possibilities:

$\{5,6,8,9\}$	$\{4,6,7,9\}$	$\{4,5,7,8\}$
$\{2,3,8,9\}$	$\{1,3,7,9\}$	$\{1,2,7,8\}$
$\{2,3,5,6\}$	$\{1,3,4,6\}$	$\{1,2,4,5\}$

Table 1: B2 Entry Possibilities

We first form an incidence matrix for these 9 subsets. Define the subsets corresponding to the cells as follows:

$$X_1 = \{5,6,8,9\}, \qquad X_2 = \{4,6,7,9\}, \qquad X_3 = \{4,5,7,8\}$$
$$X_4 = \{2,3,8,9\}, \qquad X_5 = \{1,3,7,9\}, \qquad X_6 = \{1,2,7,8\}$$
$$X_7 = \{2,3,5,6\}, \qquad X_8 = \{1,3,4,6\}, \qquad X_9 = \{1,2,4,5\}$$

Construct the matrix A_1:

$$A_1 = (a_{ij}) = \begin{cases} 1 & \text{if} x_j \in X_i \\ 0 & \text{if} x_j \notin X_i \end{cases}$$

$$A_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Proposition 2.2. For a regular $3 \times 3 \times 3$ subcube with block B_1 fixed in standard form, the incidence matrix A_1 of B_2 is regular, satisfying $A_1 \in \Lambda_9^4$. Furthermore, B_3 is completely determined by our choice for B_2.

Proof. For any empty cell (r,c) in B_2, the starting candidate pool is $3^2 = 9$. The corresponding row and column in B_1 restrict 3 digits each. To determine the overlap between these two restriction pools, we observe that the 3 digits in the row pool occupy 3 distinct columns in B_1. Because the slice rules distribute symbols uniformly, these 3 digits are partitioned equally across the 3 columns of the grid. Thus, only $\frac{3}{3} = 1$ digit overlaps with our target column c.

By the Principle of Inclusion-Exclusion, the number of excluded candidates is $3+3-1=5$. Subtracting this from the initial candidates pool leaves $9-5=4$ valid candidates per cell. Therefore, the incidence matrix A_1 for block B_2 maintains constant row and column sums of 4, meaning $A_1 \in \Lambda_9^4$.

For the final block B_3, the row and column constraints now span both B_1 and B_2. The row pool contains $2 \times 3 = 6$ restricted digits. Each of these 6 digits must appear once in each of the 2 completed blocks. Therefore, across B_1 and B_2, these 6 digits account for $2 \times 6 = 12$ total column placements. Distributing these placements uniformly across the 3 available columns gives an overlap of $\frac{12}{3} = 4$ digits shared with our target column c.

Applying inclusion-exclusion, the total excluded candidates are $6+6-4=8$. Subtracting this from the initial candidates pool leaves $9-8=1$ valid candidate per cell. This implies $A_2 \in \Lambda_9^1$. So B_3 is completely determined by our choice for B_2. □

Evaluating the incidence matrix A_1 gives $\mathrm{per}(A_1) = 448$, representing 448 valid completions for block B_2. As shown, this selection uniquely determines the final block B_3. Therefore, there are precisely 448 regular $3 \times 3 \times 3$ subcubes up to relabeling. Factoring in the 9! permutations for block B_1 gives a total of $448 \times 9! = 162,570,240$ valid regular $3 \times 3 \times 3$ subcubes.

2.3 Counting Regular Subcubes of order n

Having established the combinatorial framework in the previous section, we now generalize these structural bounds to an arbitrary order n regular subcube.

Lemma 2.3. Suppose we have m validly completed blocks $(B_1, \ldots, B_m)$ of a regular subcube. For any cell (r,c) in the subsequent block B_{m+1}, the number of available candidates drops to $(n-m)^2$. The resulting incidence matrix A_m is regular, with $A_m \in \Lambda_{n^2}^{(n-m)^2}$.

Proof. Consider any empty cell (r,c) in B_{m+1}. Our starting pool of candidates is n^2.

The row passing through this cell spans the m preceding blocks, collecting a pool of mn restricted digits. The column passing through the cell does the same, collecting another pool of mn restricted digits. To find the remaining candidates, we must subtract these restrictions using the Principle of Inclusion-Exclusion, which requires us to find the exact number of digits shared between the row pool and the column pool.

Consider the mn distinct digits in the row pool. In the m completed blocks, each of these digits has been placed m times, occupying m distinct columns. Therefore, across the m completed blocks, these mn digits account for $m \times mn = m^2 n$ total column placements.

Within the m completed blocks, each digit contributes m column placements. Due to the SSCs slice constraints, these m^2n total placements partition uniformly across the n available columns. As a result, the number of row pool digits intersecting target column c is $\frac{m^2n}{n} = m^2$.

Applying inclusion-exclusion, the total number of excluded candidates is $2mn - m^2$. Subtracting this from our initial candidates pool gives the number of remaining candidates:

$$n^2 - (2mn - m^2) = n^2 - 2mn + m^2 = (n-m)^2$$

This holds identically for every cell in B_{m+1}. The reduction applies uniformly and the incidence matrix A_m maintains constant row and column sums of $(n-m)^2$. Thus, $A_m \in \Lambda_{n^2}^{(n-m)^2}$. □

Lemma 2.4. The total number of valid completions for a regular $n \times n \times n$ subcube is given by the product of the permanents of the incidence matrices $A_0, A_1, \ldots, A_{n-1}$. The overall count is bounded by the product of Brègman's upper bounds for each $\Lambda_{n^2}^{(n-m)^2}$.

Proof. We build a regular subcube sequentially, completing B_1, then B_2, all the way through B_n. At each step m, the number of valid ways to fill the subsequent block B_{m+1} corresponds to the number of valid matchings between the available positions and the remaining entries, which is measured by the permanent of our incidence matrix, $\mathrm{per}(A_m)$.

Because each block completion builds directly on the valid structure of the ones before it, the total number of regular subcube configurations is the product of these permanents across all m steps. Applying Brègman's Theorem to the regular incidence matrices in $\Lambda_{n^2}^{(n-m)^2}$, each factor is bounded by its respective row-sum degree, giving us the upper bound for the entire space. □

We enumerate the order 4 regular subcubes by first fixing block B_1 in standard form and sequentially counting the remaining blocks. Just as in the $n = 3$ case, the slice constraints dictate that completing blocks B_1, B_2, and B_3 leaves one valid assignment for each cell in the final block, meaning B_4 is completely determined.

We construct incidence matrices to count the completions for the blocks B_2 ($m = 1$) and B_3 ($m = 2$). By Lemma 2.3, the available candidates drops quadratically at each step, dropping from 9 candidates per cell in B_2 to 4 candidates per cell in B_3. The incidence matrices A_1 and A_2 fall into the classes Λ_{16}^9 and Λ_{16}^4, respectively.

To define the bounds on the total number of configurations C, we evaluate the permanent bounds of A_1 and A_2 together with the 16! permutations for block B_1. This

gives:

$$16! \times \left(\frac{8^8}{9^7}\right)^{16} \times \left(\frac{3^3}{4^2}\right)^{16} < \mathrm{C} \leq 16! \times (9!)^{\frac{16}{9}} (4!)^{\frac{16}{4}}$$

The total number of valid configurations is bounded by:

$$4.75 \times 10^{25} < \mathrm{C} \leq 5.31 \times 10^{28}$$

We carry this exact same logic over to order 5 regular subcubes. The sequential completion of blocks B_2, B_3, and B_4 gives the incidence matrices $A_1 \in \Lambda_{25}^{16}$, $A_2 \in \Lambda_{25}^{9}$, and $A_3 \in \Lambda_{25}^{4}$:

$$25! \times \left(\frac{15^{15}}{16^{14}}\right)^{25} \times \left(\frac{8^8}{9^7}\right)^{25} \times \left(\frac{3^3}{4^2}\right)^{25} < \mathrm{C} \leq 25! \times (16!)^{\frac{25}{16}} (9!)^{\frac{25}{9}} (4!)^{\frac{25}{4}}$$

This gives the bound:

$$1.23 \times 10^{64} < \mathrm{C} \leq 1.19 \times 10^{70}$$

More generally, for regular subcubes of order $n \geq 4$, the sequential completion of blocks B_2 through B_{n-1} is governed by incidence matrices $A_m \in \Lambda_{n^2}^{k_m}$, where the row-sum degree is defined by $k_m = (n-m)^2$ for $m = 1, \ldots, n-2$. Finding the permanents of these matrices gives the general bound:

$$(n^2)! \prod_{m=1}^{n-2} \left(\frac{(k_m - 1)^{k_m - 1}}{k_m^{k_m - 2}}\right)^{n^2} < \mathrm{C} \leq (n^2)! \prod_{m=1}^{n-2} (k_m!)^{\frac{n^2}{k_m}}$$

3 Bounds for Arbitrary $x \times y \times z$ Subcubes

3.1 Matrix Augmentation for Arbitrary Subcubes

To evaluate an arbitrary $x \times y \times z$ subcube, we map its geometry by slicing it into z distinct planar cuts of dimensions $x \times y \times 1$, where each cut is within a sequential block B_m.

In Section 2, the full $n \times n$ blocks aligned perfectly with the Sudoku slice constraints. This ensured that the available candidate pool shrank uniformly across every cell in the block, keeping the incidence matrices within the class $\Lambda_{n^2}^{k}$. For an irregular $x \times y \times 1$ planar cut, this uniformity is lost.

Let $V = x \cdot y$ be the number of physical cells in the slice. Because these V cells only partially intersect the row and column restrictions left by earlier blocks, the number of

valid candidates remaining, denoted d_i, varies from cell to cell. As a result, the incidence matrix A for this cut is a rectangular $V \times n^2$ matrix with varying row sums. Because A is neither regular nor square, we cannot apply Brègman's bound directly. To resolve this, we augment the matrix using virtual rows. By appending $n^2 - V$ virtual rows of all ones to A, we construct a square $n^2 \times n^2$ augmented matrix, $\bar{A}$. These virtual rows act as unrestricted dummy cells that absorb whichever digits the physical cells do not select.

Finally, applying Lemma 1.1, we divide the result by $(n^2 - V)!$ to factor out the permutations created by the dummy rows. This step allows us to establish an upper bound for the irregular subcubes.

3.2 Generalized Upper Bound

Having constructed the square augmented matrix $\bar{A}$, we can apply the Brègman-Minc theorem (Theorem 1.3) to establish an upper bound on its permanent.

Let $N = n^2$ be the total number of digits in the SSC. For the cells in the subcube, the row sums in the incidence matrix correspond to the remaining valid candidates, $d_1, d_2, \ldots, d_V$.

For the $N - V$ virtual rows we added, every row sum is N. The Brègman-Minc bound for the augmented matrix is the product of the individual bounds for all N rows:

$$per(\bar{A}) \leq \left(\prod_{i=1}^{V} (d_i!)^{\frac{1}{d_i}} \right) \left((N!)^{\frac{1}{N}} \right)^{N-V}$$

Applying Lemma 1.1 and dividing by the permutations of the dummy rows, $(N-V)!$, we obtain the general bound for any $x \times y \times 1$ planar cut:

$$\mathrm{C} \leq \frac{1}{(N-V)!} \left(\prod_{i=1}^{V} (d_i!)^{\frac{1}{d_i}} \right) \left((N!)^{\frac{1}{N}} \right)^{N-V}$$

Where C denotes the total number of valid configurations.

To find the maximum number of valid configurations for the entire $x \times y \times z$ subcube, we evaluate this formula for each of its z slices. Multiplying these z sequential slice bounds together gives the total upper bound for the arbitrary subcube.

Example 3.1. To see how this works in practice, let's evaluate a $1 \times 2 \times 1$ arbitrary subcube located entirely within the second slice (B_2) of a regular $3 \times 3 \times 3$ subcube. Because this is embedded within an order-3 SSC, the overarching candidate pool is $N = 3^2 = 9$.

Our planar cut consists of $V = 1 \times 2 = 2$ cells. Based on the calculations established in Section 2, completing the first block B_1 restricts the available candidate pool for B_2 such that every cell has 4 valid candidates remaining. Therefore, our row sums for the cells are $d_1 = 4$ and $d_2 = 4$.

Because $V = 2$, we need to add $N - V = 7$ virtual rows to square the matrix. Each of these dummy rows will have a sum of 9. We can now plug these values directly into the general bound:

$$\mathrm{C} \leq \frac{1}{7!} \left((4!)^{\frac{1}{4}} \times (4!)^{\frac{1}{4}} \right) \left((9!)^{\frac{1}{9}} \right)^7$$

First, we calculate the bounds for the two sections of the augmented matrix:

- The cell product evaluates to $(24)^{\frac{1}{4}} \times (24)^{\frac{1}{4}} = \sqrt{24} \approx 4.899$.
- The virtual cell product evaluates to $\left((362,880)^{\frac{1}{9}} \right)^7 \approx 30,971.6$.

Multiplying these together gives the maximum permanent for the 9×9 augmented matrix, which is approximately $151,730$.

Finally, we isolate the bound for our actual cells by dividing by the permutations of the 7 dummy rows ($7! = 5,040$):

$$\mathrm{C} \leq \frac{151,730}{5,040} \approx 30.1$$

Since the number of valid completions must be an integer, there are at most 30 valid ways to complete this specific $1 \times 2 \times 1$ arbitrary subcube.

4 Future Work

These regular subcube structures can be embedded onto a physical n-Rubik's cube. This approach differs from standard novelty Sudoku cubes in that the spatial center is treated mathematically as an ordinary node. These mappings can be explored to determine the number of possible solvable puzzle states a given regular subcube can generate.

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